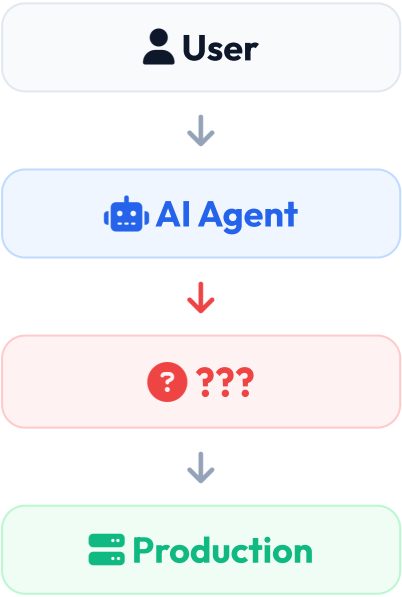
AI EVALUATION

Everyone is Building AI Agents...

But How Do You Know They Actually Work?



Check caption for the MY full research paper link

THE PROBLEM

Most Agent Evaluations Are Broken

TODAY WE MEASURE:

- Accuracy
- BLEU
- ROUGE
- Task Success

But AI Agents do much more. They:

Plan	Use Tools	Memory
Decisions	Collaborate	Adapt

Traditional metrics miss all of this.

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CASE STUDY

Why Traditional Evaluation Fails

User:

"Book me the cheapest flight to Delhi."

AGENT EXECUTION:

- Uses wrong website
- Makes 5 unnecessary tool calls
- Takes 2 minutes
- ✓ Finally books flight

Task completed?	Good agent?
YES	NO

The process matters as much as the outcome.

Based on the paper's discussion of runtime uncertainty, tool failures, and reasoning paths.

THE FRAMEWORK

What Should We Evaluate Instead?

The Agent Evaluation Framework



This paper proposes evaluating agents across all these dimensions.

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DIMENSION 1

1. Reasoning Evaluation

QUESTION:

Can the agent think correctly?

MEASURE:

- ✓ Instruction Following
- ✓ Hallucination Rate
- ✓ Goal Alignment

EXAMPLE FLOW:

Question → Reasoning → Decision → Action

Bad reasoning = bad actions.
Even if the final answer looks correct.

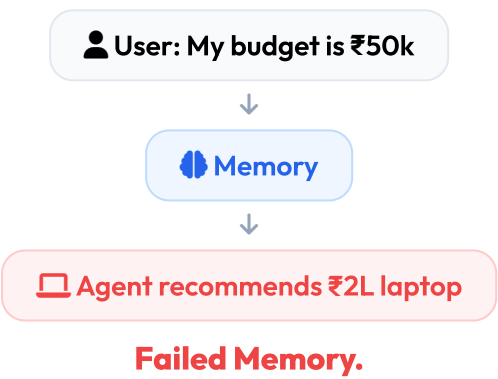
DIMENSION 2

2. Memory Evaluation

QUESTION:

Does the agent remember the right things?

EXAMPLE:



Failed Memory.

EVALUATE:

- Retrieval Accuracy
- Context Consistency
- Temporal Understanding

The paper highlights memory retrieval and consistency as critical evaluation areas.

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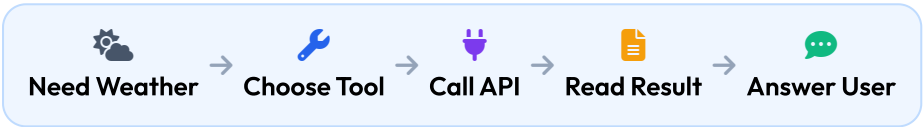
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DIMENSION 3

3. Tool Use Evaluation

Most agents fail here.

EXAMPLE FLOW:



MEASURE:

- ✓ Correct Tool Selection
- ✓ Correct Parameters
- ✓ Correct Execution Order

A smart model can still be a bad agent if it uses tools poorly.

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DIMENSION 4

4. Multi-Agent Evaluation

Modern systems use teams of agents.

EXAMPLE:



QUESTIONS:

- Did they communicate properly?
- Did they share context?
- Did they resolve conflicts?

> **Agent quality is now a team sport.**

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THE GAP

Current Benchmarks Are Not Enough

POPULAR BENCHMARKS:

- AgentBench
- SWE-bench
- ToolBench
- WebArena
- GAIA
- OSWorld

NO BENCHMARK FULLY MEASURES:

- ✕ Safety
- ✕ Memory
- ✕ Collaboration
- ✕ Real-world uncertainty

Problem: Most evaluate only parts of the system.

The paper identifies these benchmark limitations.

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CONCLUSION

My Biggest Takeaway

The future challenge isn't:

"Can we build AI Agents?"

It's:

"Can we trust them?"

AN AI AGENT SHOULD BE EVALUATED ON:

- ✓ Thinking
- ✓ Remembering
- ✓ Acting
- ✓ Using Tools
- ✓ Collaborating
- ✓ Staying Safe

A successful task does NOT always mean a successful agent.

What metric matters most for AI Agents?

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